

Quiz 2

15 minutes · closed book · no calculators

Three questions, five minutes each. Drawn from the core problems of Homeworks 3 and 4. Answers should be short; a correct one-line argument is worth more than a page of work.

1. COMPUTE Let $B \sim \text{Bernoulli}(p)$, and let $B_{n,1}, \dots, B_{n,n}$ be *independent* $\text{Bernoulli}(\lambda/n)$ with $N_n = \sum_{i \leq n} B_{n,i}$.

- (a) Write down $\varphi_B(t) = \mathbb{E}[e^{itB}]$.
- (b) Write down $\varphi_{N_n}(t)$, compute $\lim_{n \rightarrow \infty} \varphi_{N_n}(t)$, and name the limiting law.
- (c) Which property of the $B_{n,i}$ did you use to get from (a) to (b)?

2. WHICH HYPOTHESIS FAILS Let π be a uniformly random permutation of $\{1, \dots, n\}$, let $I_i = \mathbf{1}\{\pi(i) = i\}$, and let $W = \sum_{i \leq n} I_i$ be the number of fixed points.

- (a) Compute $\mathbb{E}W$.
- (b) Le Cam's theorem states that for *independent* $B_i \sim \text{Bernoulli}(p_i)$ with $\lambda = \sum_i p_i$,

$$d_{\text{TV}}\left(\sum_i B_i, \text{Poisson}(\lambda)\right) \leq \sum_i p_i^2.$$

A student applies it with $p_i = 1/n$ and $\lambda = 1$ and concludes $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 1/n$. Name precisely the hypothesis that fails.

- (c) That conclusion is in fact true. Say in one sentence why this does not rescue the argument.

3. SUPPLY THE MISSING STEP Let $X_1, \dots, X_n$ be i.i.d. with mean 0 and variance 1, and set $W = n^{-1/2} \sum_{i \leq n} X_i$. Let I be uniform on $\{1, \dots, n\}$ and X'_I an independent copy of X_I , both independent of everything else, and put

$$W' = W - \frac{X_I - X'_I}{\sqrt{n}}.$$

- (a) Compute $\mathbb{E}[W' - W \mid X_1, \dots, X_n]$.
- (b) Deduce the value of a in the Stein-pair relation $\mathbb{E}[W' - W \mid W] = -aW$.
- (c) In one sentence: why is a pair like this useful when the summands are *not* independent?

Name: _____